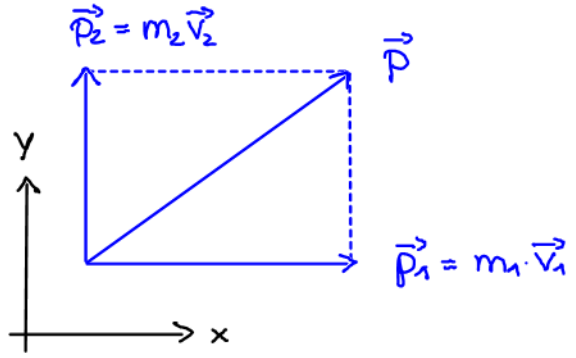
Lösung: Mehrdim. inelastischer Stoß bei Autounfall

a) Impulserhaltung (in 2D):



"vorher"		"nachher"
$\vec{p}_1 + \vec{p}_2$	=	$\vec{p}$
$m_1\vec{v}_1 + m_2\vec{v}_2$	=	$(m_1+m_2)\vec{v}$

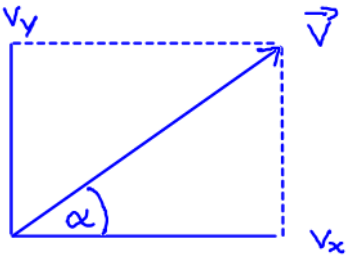
$$m_1\begin{pmatrix} v_1 \\ 0 \end{pmatrix} + m_2\begin{pmatrix} 0 \\ v_2 \end{pmatrix} = (m_1+m_2)$$

$$\vec{v} = \begin{pmatrix} v_x \\ v_y \end{pmatrix} = \frac{1}{m_1+m_2}\begin{pmatrix} m_1 v_1 \\ m_2 v_2 \end{pmatrix} = \begin{pmatrix} \frac{m_1 v_1}{m_1+m_2} \\ \frac{m_2 v_2}{m_1+m_2} \end{pmatrix}$$

$$= \begin{pmatrix} \frac{1000\,kg \cdot 27{,}78\,m/s}{2200\,kg} \\ \frac{1200\,kg \cdot 11{,}11\,m/s}{2200\,kg} \end{pmatrix} = \begin{pmatrix} 12{,}63\,m/s \\ 6{,}06\,m/s \end{pmatrix} = \begin{pmatrix} 45{,}45\,km/h \\ 21{,}82\,km/h \end{pmatrix}$$

$$|\vec{v}| = \sqrt{v_x^2 + v_y^2} \approx \sqrt{(12{,}63\,m/s)^2 + (6{,}06\,m/s)^2} \approx \underline{\underline{14{,}0\,m/s}} \quad (= 50{,}4\,km/h)$$

$$\tan(\alpha) = \frac{v_y}{v_x} = \frac{\frac{m_2 v_2}{m_1+m_2}}{\frac{m_1 v_1}{m_1+m_2}}$$



$$\alpha = \arctan\left(\frac{m_2 v_2}{m_1 v_1}\right)$$

$$\approx \arctan\left(\frac{1200\,kg \cdot 11{,}11\,m/s}{1000\,kg \cdot 27{,}78\,m/s}\right)$$

$$\underline{\underline{\alpha \approx 25{,}63°}}$$

Verformungs- & Wärmeenergie

b) $E_{kin,ges}^{(vorher)} = E_{kin,ges}^{(nachher)} + \Delta Q$

$$\Delta Q = E_{kin,ges}^{(vorher)} - E_{kin,ges}^{(nachher)}$$

$$= \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2 - \frac{1}{2}(m_1+m_2) v^2$$

$$= \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2 - \frac{1}{2}(m_1+m_2)\left(\frac{1}{m_1+m_2}\sqrt{(m_1 v_1)^2+(m_2 v_2)^2}\right)^2$$

$$= \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2 - \frac{1}{2}\frac{(m_1 v_1)^2+(m_2 v_2)^2}{m_1+m_2}$$

$$= \frac{1}{2}\left(m_1 - \frac{m_1^2}{m_1+m_2}\right) v_1^2 + \frac{1}{2}\left(m_2 - \frac{m_2^2}{m_1+m_2}\right) v_2^2$$

$$= \frac{1}{2} m_1 \left(1 - \frac{m_1}{m_1+m_2}\right) v_1^2 + \frac{1}{2} m_2 \left(1 - \frac{m_2}{m_1+m_2}\right) v_2^2$$

$$= \frac{1}{2}\frac{m_1 m_2}{m_1+m_2} v_1^2 + \frac{1}{2}\frac{m_1 m_2}{m_1+m_2} v_2^2 = \underline{\underline{\frac{1}{2}\mu_R (v_1^2+v_2^2)}}$$

$$\Delta Q = \frac{1}{2}\cdot\frac{1000\,kg\cdot 1200\,kg}{2200\,kg}\left((27{,}78\,m/s)^2+(11{,}11\,m/s)^2\right)$$

$$\underline{\underline{\Delta Q \approx 244134{,}7\,J \approx 2{,}4\cdot 10^5\,J}}$$

c) $$P = \frac{\Delta Q}{E_{kin,ges}^{(vorher)}} = \frac{\frac{1}{2}\mu_R (v_1^2+v_2^2)}{\frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2}$$

$$\approx \frac{244134{,}7\,J}{\frac{1}{2}\,1000\,kg\cdot(27{,}78\,m/s)^2 + \frac{1}{2}\,1200\,kg\cdot(11{,}11\,m/s)^2}$$

$$\approx \frac{244134{,}7\,J}{459923{,}46\,J} \approx 0{,}531 \approx \underline{\underline{53{,}1\,\%}}$$